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A Pattern Reconfigurable MIMO Antenna Design Using Characteristic Modes

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ABSTRACT A novel pattern-reconfigurable two-port multiple-input multiple-output (MIMO) antenna system at Wi-Fi band has been designed by resorting to characteristic modes analysis. The design method of the proposed MIMO antenna array is based on the selective excitation of three orthogonal characteristic modes. The selective excitation is achieved by integrating two pairs of capacitive coupling elements (CCEs) on the same plane with the ground. Two feeding networks related to two ports have been designed for achieving a good impedance matching and providing the feeding to the CCEs. By switching the p-i-n diodes in each feeding network, the selective excitation mode of the two ports can be varied, thus leading to the reconfigurable patterns. Three operating states can be obtained for the proposed antenna design. Simulation results demonstrate that the envelope correlation coefficients of the proposed MIMO antenna array are far less than 0.5 due to the orthogonality of radiation patterns between the characteristic modes in each of the three states. Measured S-parameters and the radiation patterns of a realized prototype are in good agreement with simulation ones confirming the reliability of the design method.

INDEX TERMS Characteristic modes, radiation-reconfigurable, MIMO system, orthogonality of radiation patterns.

I. INTRODUCTION

Due to growing demands from users to access higher data rates in the advanced mobile communication, multi-input multi-output (MIMO) technology has attracted increasing attentions. With multiple antennas in both base stations and terminals, higher data rates and larger channel capacity can be obtained via signal multipath propagation [1]–[3]. However, the wireless devices in real life usually operate in a dynamic environment. When operating in an environment where there are a few paths that a single can take from the transmitter to the receiver, the capacity of the MIMO system consisting of fixed-pattern antennas will be affected. One possible solution is to use the MIMO system with radiation pattern diversity.

Recently, the pattern-reconfigurable MIMO antennas have been developed [4]–[10]. In [4], the antenna with two switches is able to reconfigure the radiation patterns in three different directions. Two antennas are orthogonally located on the chassis, thus achieving good isolation. A compact pattern reconfigurable antenna consisting of a U-slot patch and eight shorting posts was designed in [5]. With switching

the PIN diodes, the radiation patterns of the antenna can be changed between conical and boresight patterns. By enlarging the space of the two antennas to be one wavelength, the low mutual coupling is obtained. A two-layer antenna structure with conical and boresight patterns was proposed in [6]. Two decoupling methods, i.e., decoupling slits and shielding wall, are located between the two antennas with a separation of 0.25λ to improve the isolation. In the abovementioned cases, the MIMO system design usually is based on integrating multiple same antenna elements, and some decoupling techniques are employed to achieve the low mutual coupling correlation.

Recently, characteristic modes analysis (CMA) has been developed in the design of the MIMO antenna systems [11]–[22]. In the CMA, the ground acts as a main radiator, and the antenna design is performed by exciting the desired modes of the ground. There are two widely used approaches to excite the specific characteristic mode, i.e., inductive coupling method and capacitive coupling method. Generally, the optimal location of the inductive exciter is

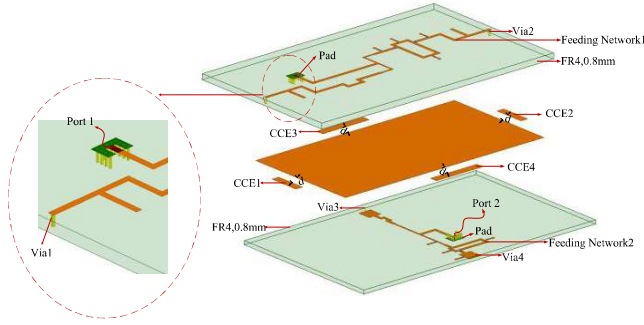


FIGURE 1. Geometry of the proposed pattern reconfigurable antenna.

at the maximum characteristic current, while the capacitive one is at the maximum electric field [23]. Due to the nature orthogonality properties of the characteristic modes, good isolation between the antenna elements can be easily achieved. Recently, a pattern-reconfigurable MIMO antenna with two tunable parasitic elements was developed [24]. By connecting or disconnecting the parasitic elements with the chassis, the different characteristic modes can be generated to provide pattern diversity. In [25] and [26], antennas with reconfigurable patterns were designed by selectively exciting different characteristic modes. The phase shifter was introduced to control the excitation degrees of the desired modes for achieving the null-scanning of the radiation pattern.

In this paper, a pattern-reconfigurable two-port MIMO antenna system is designed at WIFI band. According to the CMA, design of the antenna element with pattern diversity is based on selective excitation of different characteristic modes, which is similar to those in [25] and [26]. To realize the MIMO antenna array, a design method for simultaneous excitation of two reconfigurable orthogonal modes has been developed, which is main contribution of this paper. In the proposed MIMO antenna system, two pairs of capacitive coupling elements (CCEs) are integrated with the ground, and two feeding networks related to the two ports have been designed. By switching the states of the p-i-n diodes, the feeding networks can generate three kinds of the feeding structures, thus leading to three possible states for this reconfigurable antenna. In each state of the antenna system, the two ports can simultaneously excite two orthogonal mode patterns, thus resulting in good isolation of the proposed MIMO system.

II. PATTERN RECONFIGURABLE MIMO ANTENNA DESIGN

The two-port pattern-reconfigurable MIMO antenna system is depicted in Fig. 1. The antenna consists of three layers: a feeding network 1, a common metallic ground with a size of 120 mm×60 mm and a feeding network 2. Between the three layers, two 0.8 mm-thick FR4 substrates with a permittivity of 4.2 and a loss tangent of 0.02 are used. The specific structures of the two feeding networks will be described in the following section. In the proposed MIMO antenna system,

the ground plane acts as a main radiator and two pairs of the CCEs are located on the same layer with the ground for exciting the current on it. The CCE 1 has a same size of 4 mm×16 mm as the CCE 2, and they are located in the middle of two shorter edges of the ground, respectively. The CCEs 3 and 4 which are located in the middle of two longer edges, respectively, have a same size of 28 mm×4 mm. The distance d between the CCE and the ground is set as 4mm. As shown in Fig.1, the two feeding networks are connected with two pairs of the CCEs through four metallic vias (vias 1, 2, 3 and 4), respectively. Two same pads connected with the metallic ground and UFL connectors are used to feed the two ports.

To achieve the proposed antenna design, two techniques have been used, i.e., the CMA and the reconfigurable feeding network. The CMA is used to analyze the operating principle of the ground and selective excitation of the desirable modes, and the reconfigurable feeding network is designed to achieve a good impedance matching and generate three kinds of feeding structures, thus leading to three possible states for the pattern-reconfigurable MIMO antenna.

A. CHARACTERISTIC MODES ANALYSIS

The CMA starts from the following weighted eigenvalue equation [15], [27], i.e.,

$$X(J_n) = \lambda_n R(J_n), \quad (1)$$

in which J_n represents the eigenfunction or the characteristic current, R and X are the real and imaginary parts of the impedance matrix, respectively, and λ_n denotes the eigenvalues. The magnitude of λ_n gives information about how well the n -th mode can be excited. The smaller the magnitude of λ_n is, the more efficiently the n th mode radiates. Especially, the n th mode resonates when λ_n equals zero.

The total current J on the surface of the conducting body is expressed as a linear superposition of the characteristic current J_n as

$$J = \sum_n \alpha_n J_n, \quad (2)$$

where the modal weighting coefficient (MWC) α_n can be calculated as

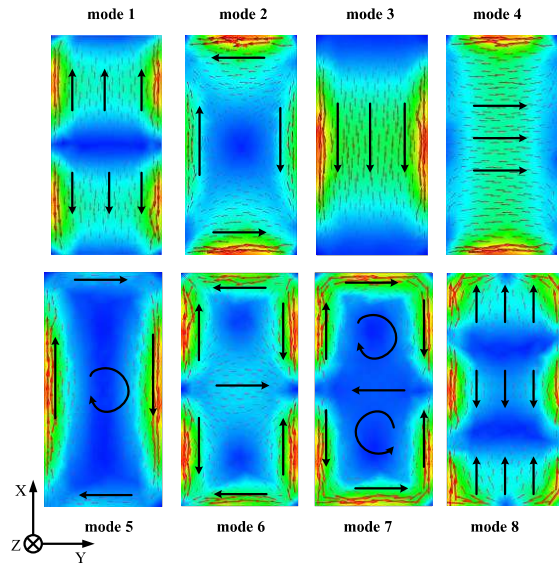
$$\alpha_n = \frac{V_n^i}{1 + j\lambda_n}. \quad (3)$$

Here the modal-excitation coefficient V_n^i depends on position, magnitude, and phase of the applied excitation. The magnitude of α_n can give the information about if a particular mode contributes largely to the total current density or not when a specific excitation is applied.

In the proposed antenna design, the weighted eigenvalue equation (1) for a 120 mm × 60 mm rectangular metallic ground at 2.45GHz is first formulated, and the corresponding eigenvalues λ_n are solved. Table 1 shows the mode eigenvalues, which correspond to the dominant radiating modes of the ground plane. It can be seen that there are 8 modes

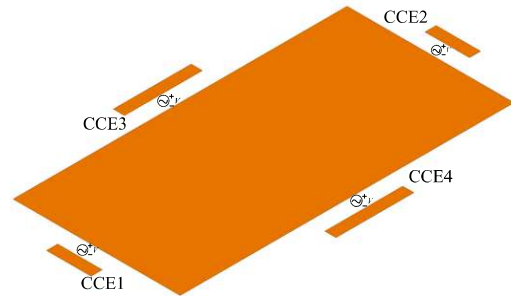
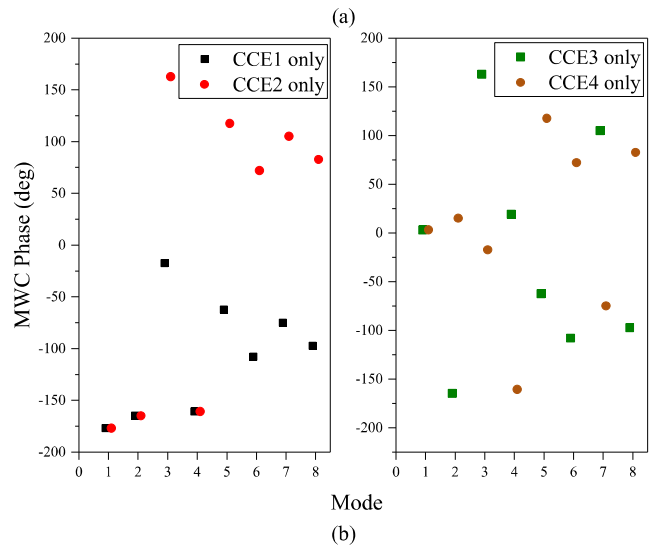
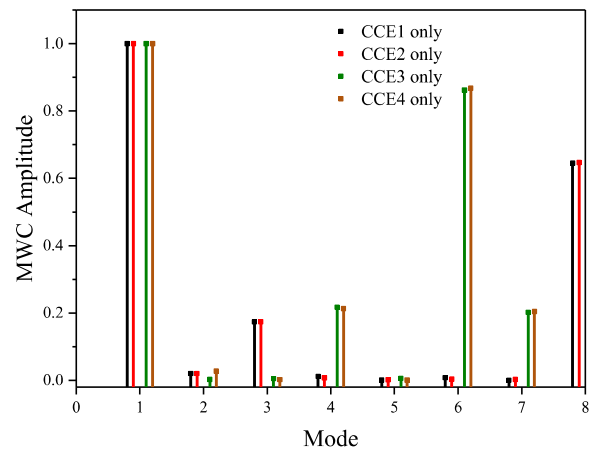
TABLE 1. Eigenvalues of dominant modes at 2.45 GHz.

Mode1	Mode2	Mode3	Mode4
-0.09	-0.26	0.32	-0.36
Mode5	Mode6	Mode7	Mode8
1.97	-3.34	3.81	-8.62

**FIGURE 2.** Characteristic current distributions of the first eight modes.

with eigenvalues near zero, and therefore they can be more easily excited. Fig. 2 shows the current distributions of these eight modes. It can be seen from Fig. 2 that the characteristic currents J_1 , J_3 and J_8 are of full-wavelength, half-wavelength, and one-and-a-half-wavelength current distributions along x direction, respectively, while the characteristic current J_4 is of half-wavelength current distribution along y direction. The characteristic currents J_2 and J_6 have half-wavelength current distributions along y direction, and half-wavelength and full-wavelength current distributions along x direction, respectively. In addition, the characteristic currents J_5 and J_7 have special inductive nature due to their currents forming closed loops over the ground [27], which means it is more difficult to excite them compared with other modes. Either the inductive coupling method or capacitive coupling method can be employed to excite these modes. In [28], the modes 1, 3 and 4 were excited by the inductive coupling elements (ICEs). In this paper, the CCEs are used to excite the modes 1, 6, and 8.

In order to excite the three modes of the ground, two pairs of the CCEs are introduced and a voltage source is applied between the ground and the CCE, as shown in Fig. 3. The CCEs 1 and 2 with the size of $4\text{ mm} \times 16\text{ mm}$ are located in the middle of two shorter edges, respectively, because there are maximal electric fields of the modes 1 and 8. Similarly, the CCEs 3 and 4 with the size of $28\text{ mm} \times 4\text{ mm}$ are located in the middle of two longer edges, respectively, where the maximal electric fields of the modes 1 and 6 distribute.

**FIGURE 3.** 3D view of conceptual design.**FIGURE 4.** Magnitude and phase of the MWC when each CCE is excited: (a) Magnitude; (b) Phase.

In order to efficiently excite the three modes, magnitude and phase of the applied excitation besides the feeding position have an impact on the modal-excitation coefficient V_n^i according to (3). Fig. 4 depicts the magnitude and phase of the MWC α , when each CCE is driven by a voltage with the same magnitude and phase at 2.45GHz. It can be seen from Fig. 4(a) that when either the CCE 1 or the CCE 2 is driven, α_1 and α_8 are dominant and have same magnitude.

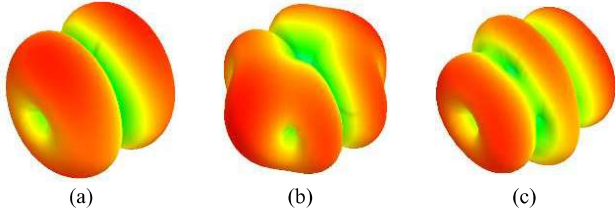


FIGURE 5. Radiation patterns of the three modes: (a) Mode 1; (b) Mode 6; (c) Mode 8.

The individual excitation of the CCE 3 or the CCE 4 results in the modes 1 and 6 being dominant and the same magnitudes of α_1 and α_6 . According to Fig. 4(b), we can know that when the CCE 1 is driven, the phases of α_1 and α_8 are -175° and -98° , respectively. With sole excitation of the CCE 2, the phases of α_1 and α_8 become -175° and 82° , respectively. This means that there is a same phase for the mode 1 and a phase difference of 180° for the mode 8 when the CCE 1 and the CCE 2 are individually driven.

Furthermore, according to (2), we can know that when the CCE 1 is individually driven, the characteristic current J_{CCE1} can be expressed as

$$\begin{aligned} J_{CCE1} &= \sum_n \alpha_n J_n \approx \alpha_1 J_1 + \alpha_8 J_8 \\ &= |\alpha_1| e^{j\theta_1} J_1 + |\alpha_8| e^{j\theta_8} J_8. \end{aligned} \quad (4)$$

Similarly, when the CCE 2 is individually driven, the characteristic current J_{CCE2} is obtained as

$$\begin{aligned} J_{CCE2} &= \sum_n \alpha'_n J_n \approx \alpha'_1 J_1 + \alpha'_8 J_8 \\ &= |\alpha'_1| e^{j\theta'_1} J_1 + |\alpha'_8| e^{j\theta'_8} J_8. \end{aligned} \quad (5)$$

If the CCEs 1 and 2 are driven simultaneously with the respective excitation coefficients of $a e^{j\psi}$ and $b e^{j\zeta}$, the total current becomes

$$\begin{aligned} J_{total} &\approx a e^{j\psi} J_{CCE1} + b e^{j\zeta} J_{CCE2} \\ &= (a |\alpha_1| e^{j(\theta_1+\psi)} + b |\alpha'_1| e^{j(\theta'_1+\zeta)}) J_1 \\ &\quad + (a |\alpha_8| e^{j(\theta_1+\psi)} + b |\alpha'_8| e^{j(\theta'_8+\zeta)}) J_8. \end{aligned} \quad (6)$$

According to Fig. 4, we have $|\alpha_1| = |\alpha'_1|$, $|\alpha_8| = |\alpha'_8|$, $\theta_1 = \theta'_1$, and $\theta_8 = \theta'_8 + \pi$. Therefore, (6) can be rewritten as

$$\begin{aligned} J_{total} &\approx (a |\alpha_1| e^{j(\theta_1+\psi)} + b |\alpha_1| e^{j(\theta_1+\zeta)}) J_1 \\ &\quad + (a |\alpha_8| e^{j(\theta_1+\psi)} - b |\alpha_8| e^{j(\theta_1+\zeta)}) J_8. \end{aligned} \quad (7)$$

It can be seen from (7) that when the same magnitude and phase of the excitation coefficients for the CCEs 1 and 2, i.e., $a = b$ and $\psi = \zeta$, are used, the total current becomes the characteristic current of the mode 1, J_1 . Similarly, when the same magnitude and a 180° out of phase of the excitation coefficients for the CCEs 1 and 2 are employed, the total current is the characteristic current of the mode 8, J_8 . Following a similar procedure, we can know that the current of the mode 6 J_6 is generated with the excitation of the CCEs

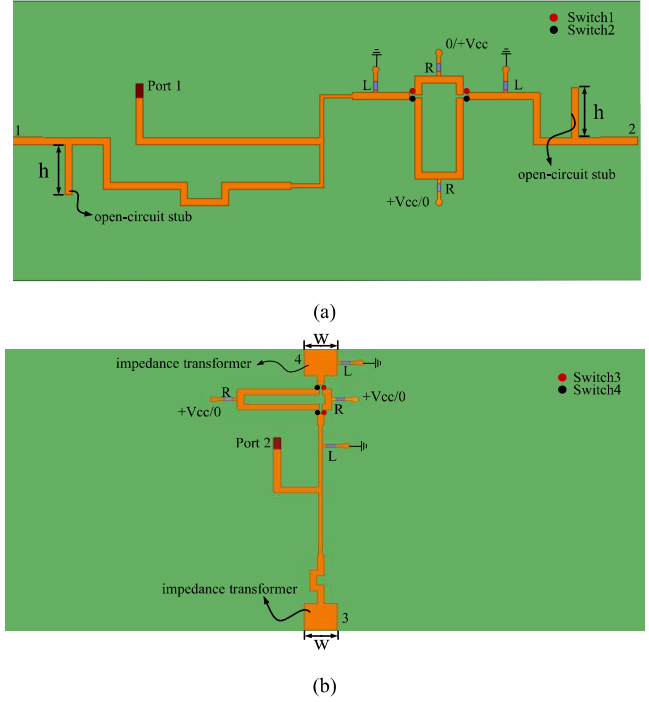


FIGURE 6. Layout of the feeding network of the two ports: (a) Port 1; (b) Port 2.

3 and 4 in the same magnitude and a phase difference of 180° , and the current of the mode 1 J_1 can be also obtained with the excitation of the CCEs 3 and 4 in the same magnitude and phase. Fig. 5 shows the characteristic patterns of the three modes when the above excitations are implemented. It can be observed that three characteristic patterns corresponding to three modes are perfectly orthogonal.

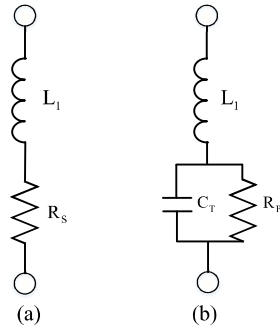
B. FEEDING NETWORK DESIGN

To achieve the proposed two-port radiation-reconfigurable MIMO antenna system, the feeding networks associated to the two ports are designed, as shown in Fig. 6. It can be seen that each feeding network is a three-port network, i.e., an input port and two output ports, and consists of three parts: matching network, power divider and phase shifter. All of the three parts were designed with microstrip line technology. In general, the eigenvalues and radiation patterns of the characteristic modes will be affected with the introduction of the feeding network. Hence the feeding structure is well designed to reduce its effect on the characteristic modes as much as possible.

As shown in Fig. 6, two open-circuit stubs in the feeding network 1 and two impedance transformers in the feeding network 2 are introduced for the impedance matching, respectively. The power divider is adopted to guarantee two outputs with the same magnitude in each feeding network. Four switches are inserted into each feeding network to adjust the phases at two output ports, as shown in Table 2. Specifically, when the switch 1 is in ON state and the switch 2 is in OFF state, the in-phase output signals at two output ports are

TABLE 2. States of the output phase and mode of the two ports.

	Switch1	Switch2	Switch3	Switch4	Port 1		Port 2	
State1	on	off	off	on	in phase	Mode1	180° out of phase	Mode6
State2	off	on	off	on	180° out of phase	Mode8	180° out of phase	Mode6
State3	off	on	on	off	180° out of phase	Mode8	in phase	Mode1

**FIGURE 7.** Equivalent circuit for the p-i-n diode: (a) Forward bias; (b) Reverse bias. $L_1 = 0.6\text{nH}$, $R_S = 2.1\Omega$, $C_T = 0.17\text{pF}$, $R_P = 5\text{K}\Omega$.

obtained in the feeding network 1. Two output signals with a phase difference of 180° are achieved when the switch 1 is in OFF state and the switch 2 is in ON state. Similarly, the in-phase output signals are obtained in the feeding network 2, when the switch 3 is turned on and the switch 4 is turned off. Inversely, output signals with the 180° out of phase are achieved. To implement the feeding network, p-i-n diodes, model Infineon BAR64-02, are used. Each switch consists of two p-i-n diodes. The corresponding equivalent circuit of the p-i-n diode shown in Fig.7 is used in the modelling of the switch.

Two feeding networks are fabricated on two 0.8 mm-thick FR4 substrates with the common ground plane, respectively. One of them is fabricated on the top of the first substrate, the other is fabricated on the bottom of the second substrate, as shown in Fig. 1. In each feeding network, two metallic vias located at two output ports are used to connect the CCEs to the feeding networks.

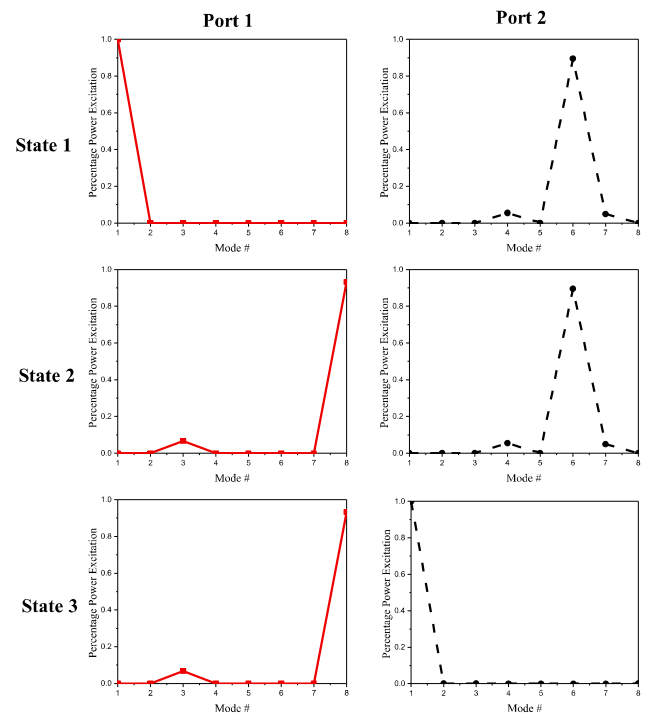
According to the previously discussed CMA, the two-port MIMO antenna system can operate in three states, as shown in Table 2. To prove the capability to achieve the desirable modes of the proposed antenna system, the percentage power excitation has been analyzed. The percentage power excitation of the n th mode p_n^{rad} can be calculated as [29]

$$p_n^{rad} = \left| a e^{j\psi} \alpha_n^{(1)} + b e^{j\zeta} \alpha_n^{(2)} \right|^2 \quad (8)$$

where $\alpha_n^{(l)}$ ($l = 1, 2$) represents the MWC associated with the l th exciter. Table 3 shows the excitation coefficients generated by the feeding networks in Fig. 6. According to Fig. 4 and Table 3, the normalized percentage power excitations of two ports in three states of the proposed MIMO antenna system are shown in Fig.8. It can be seen that the desired modes can be well excited with the proposed feeding networks.

TABLE 3. Excitation coefficients of the proposed feeding networks.

State	Amplitude (phase)			
	CCE 1	CCE 2	CCE 3	CCE 4
#1	2.46(-6.3°)	2.42(-21.1°)	5.63(85.7°)	7.60(-88.5°)
#2	3.94(0.8°)	3.06(-170.2°)	5.61(83.1°)	7.28(-89.1°)
#3	3.78(-177.7°)	3.25(0.2°)	5.51(-71.5°)	3.51(-99.5°)

**FIGURE 8.** Normalized percentage power excitations of two ports in three states.

Note that there is a weak excitation of the mode 3/mode 4 when the mode 8/mode 6 is excited. The adverse impacts of the modes 3 and 4 on the modal excitation purity can be reasonably neglected due to their small percentage power excitations.

Fig. 9 shows the radiation patterns of three states. As shown in Figs. 5 and 9, two orthogonal characteristic patterns are excited by two ports in each state. Note that in the state 1, there is a slight distortion of the pattern of the mode 1 due to the introduction of the feeding network. To evaluate the diversity performances of the proposed MIMO antenna system, the envelope correlation coefficient (ECC) at 2.45GHz is calculated using far-field radiation pattern, as shown in Table 4. The ECC far lower than 0.5 demonstrates good diversity performance in each state.

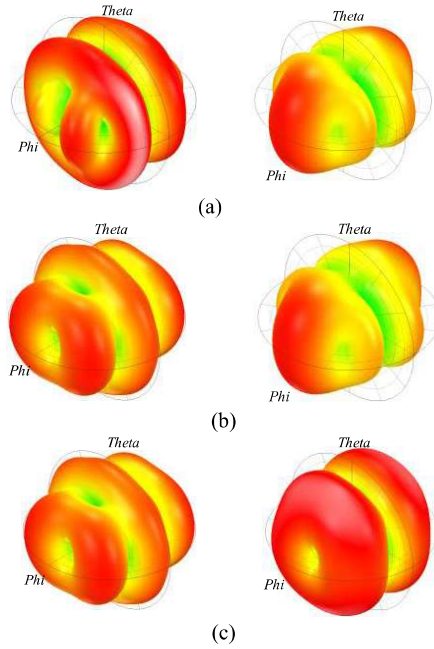


FIGURE 9. Radiation patterns from port 1 (on left) and 2 (on right) of the three states: (a) State 1; (b) State 2; (c) State 3.

TABLE 4. Envelope correlation coefficient in three states at 2.45 GHz.

State 1	State 2	State 3
0.01935	0.00045	0.00522

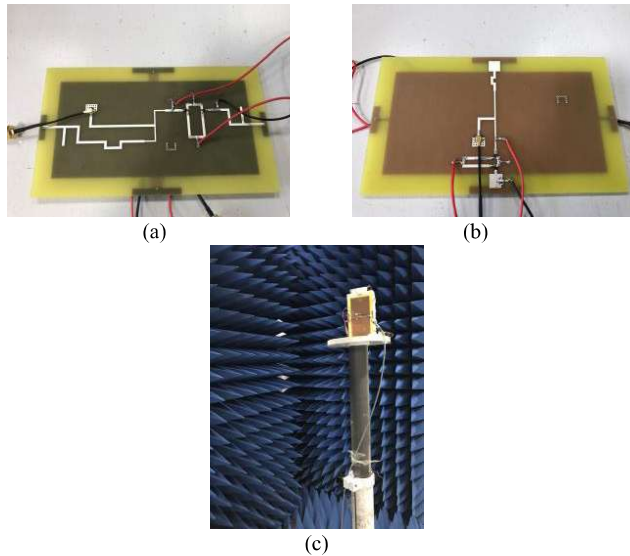


FIGURE 10. Fabricated prototype of the reconfigurable antenna: (a) Front view; (b) Rear view; (c) Testing environment.

III. EXPERIMENTAL RESULTS AND DISCUSSIONS

As shown in Fig. 10, a prototype of the radiation reconfigurable MIMO antenna array is fabricated to verify the proposed design. In the experiment, the status of the p-i-n diodes was controlled by an applied DC voltage of a 1.5-V lithium battery. In each p-i-n's bias circuit, an inductor L of 57 nH is used to blocking the RF signal from entering the DC line and a resistor R of 82 ohm is employed to set the

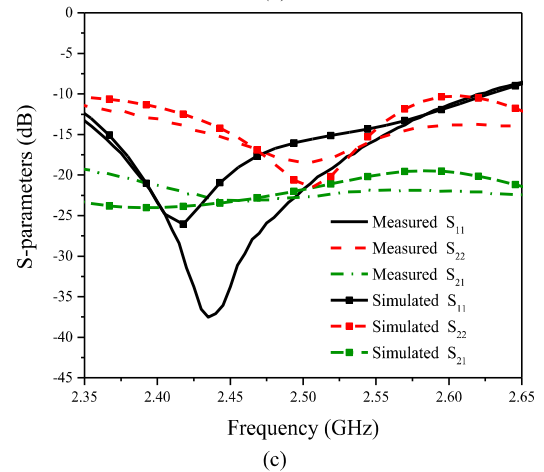
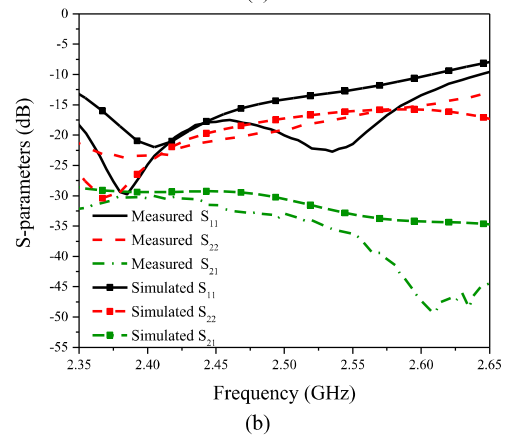
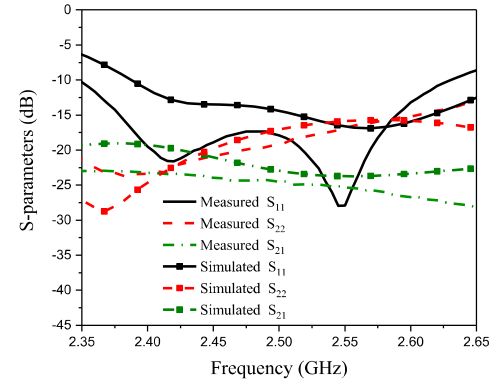


FIGURE 11. Simulated and measured S parameters of the proposed MIMO antenna array of the three states: (a) State 1; (b) State 2; (c) State 3.

current of the p-i-n diode to be around 10 mA for getting the minimum insertion loss, as shown in Fig. 6.

Fig. 11 shows measured and simulated S parameters, which are in good agreement between each other. Here the simulation was carried out by using a commercial finite-element solver Ansys HFSS. It is found that the measured operation frequency band of the three states for $S_{11} \leq -10$ dB is from 2.4 to 2.6 GHz. The measured isolation in both states 1 and 3 are better than 20dB, and the isolation is better than 30dB in the state 2. Good isolation is achieved by exciting two orthogonal modes in each of the three states.

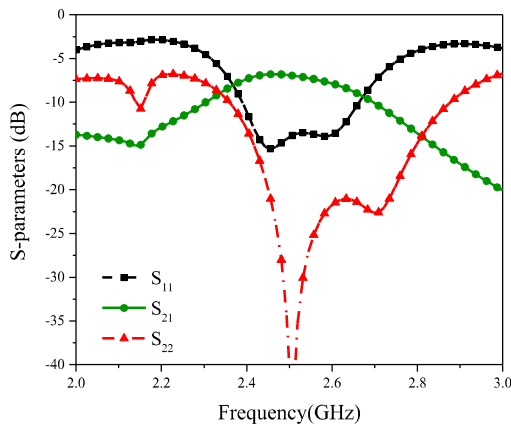


FIGURE 12. S parameters of the proposed MIMO antenna in the state 4.

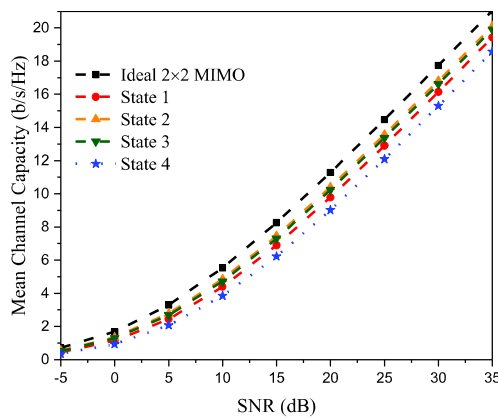


FIGURE 13. Mean channel capacity of the proposed MIMO antenna.

To further explain the reason of the good isolation of the proposed MIMO antenna array, an extra state, i.e., state 4 is considered. In the state 4, the switches 1 and 3 are in ON state while the switches 2 and 4 are in OFF state. In this case, the two ports are simultaneously exciting the mode 1. Fig. 12 shows the S parameters in the state 4. It can be seen that there is a strong coupling between the two ports due to the same radiation patterns excited by two ports.

The channel capacity of the proposed antenna system is calculated to further investigate its MIMO performance [30]. Fig. 13 shows mean channel capacity as a function of the signal-to-noise ratio (SNR) at 2.45GHz. For comparison, the mean channel capacity associated with the state 4 is also given in Fig. 13. It can be seen that the mean channel capacities of the designed antenna system in the three operating states are slightly lower than that of the ideal MIMO configuration. As the state 4 has the poor diversity performance, its mean channel capacity is lower than those in the three operating states.

Fig.14 demonstrates the simulated and measured patterns of the proposed MIMO antenna array in the three states, which are in good agreement with each other. It can be seen that the radiation patterns associated with the two ports vary as the three states are switched. In the states 1 and 2, the maximum radiation for the port 2 points in the direction

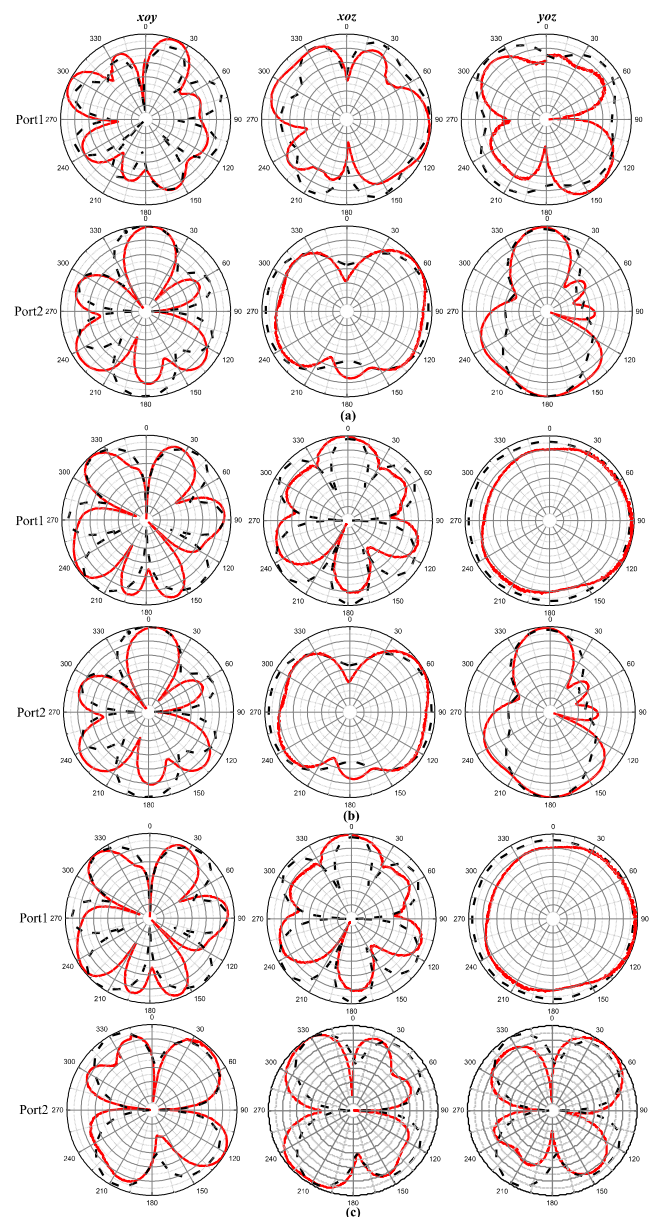


FIGURE 14. Radiation patterns (in realized gain) of the proposed MIMO antenna array at 2.45GHz of the three state (-) measured; (-) simulated: (a) State 1; (b) State 2; (c) State 3.

of $\theta = 0^\circ$ in xoy plane, while in the state 3, the maximum radiation for the port 2 points towards the direction of $\theta = 45^\circ$ in xoy plane. With the diverse radiation patterns, the proposed MIMO antenna system can be well operated in the dynamic environments. The total efficiencies of the proposed MIMO antenna array were measured. In the state 1, the total efficiencies from the ports 1 and 2 are 50% and 55%, respectively. In the state 2, these are 51% and 60%, respectively. In the state 3, these are 55% and 58%, respectively. By comparison, the simulated total efficiencies in the state 1 are 56% and 75%, when the ports 1 and 2 are excited, respectively. In the state 2, the simulated total efficiencies of the antennas 1 and 2 are 70% and 75%, respectively, and in the state 3, these are 68% and 70%, respectively. The slight discrepancy between

TABLE 5. Performance comparison between previously reported reconfigurable MIMO antennas and this work.

Ref.	Distance between antennas (λ)	Height of the antennas (λ)	S_{21} (dB)	ECC	Efficiency (%)	BW (%)
[5]	1	0.057	<-20	<0.04	45-87	6.6
[6]	0.25	0.06	<-15	<0.05	68-85	3.8
[24]	0	0.05	<-20	<0.06	71-80	7.4
This work	0	0.01	<-20	<0.02	50-60	8

the simulation and measurement is due to the fabrication errors and the diode losses. The measured total efficiencies of the two ports are larger than 40% in all of the three states, which is acceptable in practical applications.

Finally, a comparison among the reported reconfigurable MIMO antennas and this work is summarized in Table 5. With the characteristic mode design, the metallic ground acts as a radiating structure in the MIMO system, and thus the implementation complexity of the proposed work is very low. Meanwhile, due to the orthogonal property between characteristic modes, the decoupling structure is neglected, and the resulting geometry of the proposed work is compact with good ECC. On the other hand, the proposed work has a wider measured operation bandwidth and moderate measured total efficiency compared with the reported reconfigurable MIMO antennas.

IV. CONCLUSION

In this paper, a novel CMA-based two-port MIMO antenna system with the radiation pattern diversity has been designed at WIFI frequency band. According to the CMA, three characteristic modes of a metallic ground can be efficiently excited by using two pairs of the CCEs. Two reconfigurable feeding networks associated with two ports have been designed. By adjusting the states of the p-i-n diodes, two orthogonal characteristic patterns related to the two ports are switched. The simulated and measured results demonstrate good performances of the proposed pattern-reconfigurable MIMO antenna including the isolation better than 20dB, the envelope correlation coefficient far less than 0.5, and total efficiency higher than 50% across all the operating states, which is very attractive for 2.5GHz Wi-Fi applications.

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REFERENCES

- [1] M. A. Jensen and J. W. Wallace, "A review of antennas and propagation for MIMO wireless communications," *IEEE Trans. Antennas Propag.*, vol. 52, no. 11, pp. 2810–2824, Nov. 2004.
- [2] K. Li, Y. Shi, and C.-H. Liang, "Quad-element multi-band antenna array in the smart mobile phone for LTE MIMO operations," *Microw. Opt. Technol. Lett.*, vol. 58, no. 11, pp. 2619–2626, 2016.

- [3] Y. Shi, L. Zhang, and C.-H. Liang, "Dual zeroth-order resonant USB dongle antennas for 4G MIMO wireless communications," *Int. J. Antennas Propag.*, vol. 2015, Mar. 2015, Art. no. 816051.
- [4] C. Rhee *et al.*, "Pattern-reconfigurable MIMO antenna for high isolation and low correlation," *IEEE Antennas Wireless Propag. Lett.*, vol. 13, pp. 1373–1376, 2014.
- [5] P.-Y. Qin, Y. J. Guo, A. R. Weily, and C.-H. Liang, "A pattern reconfigurable U-slot antenna and its applications in MIMO systems," *IEEE Trans. Antennas Propag.*, vol. 60, no. 2, pp. 516–528, Feb. 2012.
- [6] H. Li, B. K. Lau, and S. He, "Design of closely packed pattern reconfigurable antenna array for MIMO terminals," *IEEE Trans. Antennas Propag.*, vol. 65, no. 9, pp. 4891–4896, Sep. 2017.
- [7] N. H. Chamok, M. H. Yilmaz, A. Arslan, and M. Ali, "High-gain pattern reconfigurable MIMO antenna array for wireless handheld terminals," *IEEE Trans. Antennas Propag.*, vol. 64, no. 10, pp. 4306–4315, Oct. 2016.
- [8] K. K. Kishor and S. V. Hum, "A two-port chassis-mode MIMO antenna," *IEEE Antennas Wireless Propag. Lett.*, vol. 12, pp. 690–693, 2013.
- [9] D. Piazza, M. Capacchione, J. Kountouriotis, M. D'Amico, and K. R. Dandekar, "Stacked reconfigurable circular patch antenna for adaptive MIMO systems," in *Proc. Int. Conf. Electromagn. Adv. Appl. (ICEAA)*, Turin, Italy, Sep. 2009, pp. 636–639.
- [10] D. Piazza, N. J. Kirsch, A. Forenza, R. W. Heath, Jr., and K. R. Dandekar, "Design and evaluation of a reconfigurable antenna array for MIMO systems," *IEEE Trans. Antennas Propag.*, vol. 56, no. 3, pp. 869–881, Mar. 2008.
- [11] R. Martens, J. Holopainen, E. Safin, J. Ilvonen, and D. Manteuffel, "Optimal dual-antenna design in a small terminal multiantenna system," *IEEE Antennas Wireless Propag. Lett.*, vol. 12, pp. 1700–1703, 2013.
- [12] I. Szini, A. Tatomirescu, and G. F. Pedersen, "On small terminal MIMO antennas, harmonizing characteristic modes with ground plane geometry," *IEEE Trans. Antennas Propag.*, vol. 63, no. 4, pp. 1487–1497, Apr. 2015.
- [13] M. Bouezzeddine and W. L. Schroeder, "Design of a wideband, tunable four-port MIMO antenna system with high isolation based on the theory of characteristic modes," *IEEE Trans. Antennas Propag.*, vol. 64, no. 7, pp. 2679–2688, Jul. 2016.
- [14] D.-W. Kim and S. Nam, "Systematic design of 3-port bug-like MIMO antenna based on theory of characteristic mode," in *Proc. 11th Eur. Conf. Antennas Propag. (EUCAP)*, Paris, France, Mar. 2017, pp. 2224–2227.
- [15] K. Li and Y. Shi, "Wideband MIMO handset antenna design based on theory of characteristic modes," *Int. J. RF Microw. Comput.-Aided Eng.*, vol. 28, no. 4, p. e21217, 2018.
- [16] B. Yang and J. J. Adams, "Systematic shape optimization of symmetric MIMO antennas using characteristic modes," *IEEE Trans. Antennas Propag.*, vol. 64, no. 7, pp. 2668–2678, Jul. 2016.
- [17] C. Deng, Z. Feng, and S. V. Hum, "MIMO mobile handset antenna merging characteristic modes for increased bandwidth," *IEEE Trans. Antennas Propag.*, vol. 64, no. 7, pp. 2660–2667, Jul. 2016.
- [18] H. Li, Z. T. Miers, and B. K. Lau, "Design of orthogonal MIMO handset antennas based on characteristic mode manipulation at frequency bands below 1 GHz," *IEEE Trans. Antennas Propag.*, vol. 62, no. 5, pp. 2756–2766, May 2014.
- [19] Z. Miers, H. Li, and B. K. Lau, "Design of bandwidth-enhanced and multiband MIMO antennas using characteristic modes," *IEEE Antennas Wireless Propag. Lett.*, vol. 12, pp. 1696–1699, Nov. 2013.
- [20] A. Araghi and G. Dadashzadeh, "Oriented design of an antenna for MIMO applications using theory of characteristic modes," *IEEE Antennas Wireless Propag. Lett.*, vol. 11, pp. 1040–1043, 2012.
- [21] L. Qu, H. Lee, H. Shin, M.-G. Kim, and H. Kim, "MIMO antennas using controlled orthogonal characteristic modes by metal rims," *IET Microw., Antennas Propag.*, vol. 11, no. 7, pp. 1009–1015, Feb. 2017.
- [22] Z. Miers, H. Li, and B. K. Lau, "Design of bezel antennas for multiband MIMO terminals using characteristic modes," in *Proc. Eur. Conf. Antennas Propag.*, The Hague, The Netherlands, Apr. 2014, pp. 2556–2560.
- [23] R. Martens, E. Safin, and D. Manteuffel, "Inductive and capacitive excitation of the characteristic modes of small terminals," in *Proc. Loughborough Antennas Propag. Conf.*, 2011, pp. 1–4.
- [24] K. Kumar Kishor and S. V. Hum, "A pattern reconfigurable chassis-mode MIMO antenna," *IEEE Trans. Antennas Propag.*, vol. 62, no. 6, pp. 3290–3298, Jun. 2014.
- [25] F. A. Dicandia, S. Genovesi, and A. Monorchio, "Null-steering antenna design using phase-shifted characteristic modes," *IEEE Trans. Antennas Propag.*, vol. 64, no. 7, pp. 2698–2706, Jul. 2016.

- [26] F. A. Dicandia, S. Genovesi, and A. Monorchio, "Advantageous exploitation of characteristic modes analysis for the design of 3-D null-scanning antennas," *IEEE Trans. Antennas Propag.*, vol. 65, no. 8, pp. 3924–3934, Aug. 2017.
- [27] M. Cabedo-Fabres, E. Antonino-Daviu, A. Valero-Nogueira, and M. F. Batalle, "The theory of characteristic modes revisited: A contribution to the design of antennas for modern applications," *IEEE Antennas Propag. Mag.*, vol. 49, no. 5, pp. 52–68, Oct. 2007.
- [28] R. Martens and D. Manteuffel, "Systematic design method of a mobile multiple antenna system using the theory of characteristic modes," *Microw. Antennas Propag.*, vol. 8, no. 12, pp. 887–893, Sep. 2014.
- [29] F. A. Dicandia, S. Genovesi, and A. Monorchio, "Efficient excitation of characteristic modes for radiation pattern control by using a novel balanced inductive coupling element," *IEEE Trans. Antennas Propag.*, vol. 66, no. 3, pp. 1102–1113, Mar. 2018.
- [30] F. A. Dicandia, S. Genovesi, and A. Monorchio, "Analysis of the performance enhancement of MIMO systems employing circular polarization," *IEEE Trans. Antennas Propag.*, vol. 65, no. 9, pp. 4824–4835, Sep. 2017.



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